



# Cloud Infrastructure and Services

## Module I: Cloud Computing Fundamental Weightage

1. Cloud Computing Fundamental: Cloud computing definition,
  2. Characteristics of Cloud computing as per NSIT,
  3. private, public and hybrid cloud.
  4. Cloud types; IaaS, PaaS, SaaS.
  5. Benefits and challenges of cloud computing;
  6. public vs private clouds,
  7. role of virtualization in enabling the cloud,
  8. Steps involved in transitioning from Classic data center to Cloud computing environment;
  9. Business Agility: Benefits and challenges to cloud architecture.
  10. Application, availability, performance, security and disaster recovery;
  11. next generation Cloud Application.
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## **Module II: Cloud Application**

1. Cloud Application:
  2. Technologies and the processes required when deploying web services.
  3. Deploying a web service from inside and outside a cloud architecture, advantages and disadvantages.
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## **Module III: Cloud Services Management**

1. Cloud Services Management: Reliability, availability and
  2. security of services deployed from cloud.
  3. Performance and scalability of services,
  4. tools and technologies used to manage cloud services deployment;
  5. Cloud Economics: Cloud Computing infrastructures available for implementing cloud based services.
  6. Economics of choosing a Cloud platform for an organization, based on application requirements, economic constraints and business needs (e.g. Amazon, Microsoft and Google, Salesforces.com, Ubuntu and Redhat)
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## **Module IV: Case Study: Application Development**

1. Application Development: Service creation environments to develop cloud based applications.
  2. Development environments for service development; Amazon, Azure, Google App 15%
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## **Module V: Cloud Security and Migration to Cloud**

1. Security concerns and counter measures in Cloud environment,

2. Governance, Risk, and Compliance aspects in Cloud,
3. Cloud security best practices,
4. Cloud models suitable for different categories of users,
5. Considerations for choosing applications suitable for Cloud,
6. Different phases to adopt the Cloud.
7. Best Practice Cloud IT Model: Analyze of Case Studies when deciding to adopt cloud computing architecture.
8. How to decide if the cloud is right for your requirements.
9. Cloud based service,
10. applications and development platform deployment so as to improve the total cost of ownership (TCO).